

Atlas Runbook

Every task, in order, mapped to its phase — plus what the three interfaces you'll build should actually look like. The plan says why; this says do this, then this.

YOUR QUESTION

Are you collecting training data during calibration?

Not from the calibration shots themselves — but you start collecting real data two or three days in, well before the app exists.

The five phase-0 shots are throwaway. One pose, one side, taken before the rig is locked and before `RUBRIC.md` exists. They answer "does 60° work" and nothing else.

But calendar time is the one input you cannot buy back later. Sixty days of history takes sixty days, and no amount of effort in week 6 recovers a session you didn't shoot in week 1. So the order is:

- **Days 1–2** — rig up, shoot the five calibration repeats, run Face Mesh, **lock the angle**. This part is fast; it's five photos and one script.
- **Day 3 onward** — start shooting real three-pose sessions *manually*, with your normal camera app, into `data/raw/YYYY-MM-DD/`. These are real data: day 1 of your longitudinal series and the first of your training images.
- **Meanwhile** — the heavier registration spike (LightGlue, LoFTR, error measurement) and then the capture app get built while sessions accumulate in the background.

THE ONE ORDERING RULE THAT MATTERS

Don't start real sessions until the angle is locked. If phase 0 pushes you from 60° to 50°, every session shot at 60° beforehand is at the wrong angle and can't join the series. That's the only reason to wait — and it's a two-day wait, not a two-week one. Everything else about the app is automation of something you can already do by hand.

Expect roughly **5–8 manually-shot sessions** before the capture app takes over. Name the folders correctly from the first one and they flow into the pipeline untouched.

Timeline

PHASE	W1	W2	W3	W4	W5	W6	W7	W8
0 Spike & angle lock								
– Daily capture (manual)								
1 Capture rig & app								
– Daily capture (app)								
2 Labeling								
3 Canonical space								
4 Detector								
5 Lesion tracking								
6 Visualization								

Red rows are daily capture — they never stop once started. Everything else is build work that happens around them. The two rows are separate because that's the point: capture is not a phase you finish.

The checklist

PHASE 0 Spike & angle lock	WEEK 1
☐ Print an ArUco marker (4×4_50 dictionary, ~30 mm) and a small color patch on matte paper — gloss will specular and ruin the white balance reference	
☐ Mount both on a headband or a small stand that sits in frame for every pose	
☐ Set up one diffused lamp in a fixed position; blinds shut, overhead lights off	
☐ Floor tape for your feet; wall marks left and right at 60° to turn toward	
☐ Write down the setup: camera distance, lamp position and height, time of day. You will need to rebuild this exactly after any disruption	
☐ Install: <code>python</code> , <code>opencv-python</code> , <code>mediapipe</code> , <code>numpy</code> , <code>torch</code> with CUDA, <code>kornia</code> (LoFTR / LightGlue)	
☐ Shoot one pose — left 60° — five separate times over three days, setting up from scratch each time, on both the main and 2× lens	
☐ Run MediaPipe Face Mesh on all five. Does it fire every time? Are the landmarks stable across repeats?	
☒ GATE – if Face Mesh drops out, step down to 50° and re-shoot. LOCK THE ANGLE HERE.	
☐ Start shooting real 3-pose sessions manually into <code>data/raw/YYYY-MM-DD/</code> — frontal, left_60, right_60. Every day from now on, forever <code>RUNS IN PARALLEL</code>	
☐ Hand-mark 8–10 corresponding points across the five calibration images (moles, freckles, scars) as registration ground truth	
☐ Freeze one of the five as the atlas; register the other four to it — ORB/SIFT + RANSAC first, then LightGlue + DISK, then LoFTR	
☐ Compute median and p95 reprojection error in millimetres for each method	
☐ If homography residuals are structured (worse toward the edges), fit a thin-plate spline instead	

- ☐ Compare lens sharpness at the sideburn vs the nose *within one frame*. Pick a lens and a working distance
- ☒ GATE – median error < 1.5 mm and p95 < 3 mm, or reduce yaw and retry
- ☐ Write `RUBRIC.md` : minimum lesion size in mm, the elevated-or-erythematous rule, and explicit negatives, each with an example crop

- ☐ Install `mkcert`, generate a local cert, trust it on your phone — `getUserMedia` needs a secure context and `localhost` won't help you there
- ☐ FastAPI skeleton serving a static page over HTTPS on the LAN
- ☐ SQLite init: `sessions` and `images` tables
- ☐ `POST /capture` — writes the image plus a sidecar JSON with device, lens, granted constraints, timestamp
- ☐ Phone page: `getUserMedia` on the rear camera, full-resolution stills via `ImageCapture.takePhoto()` — **not** a canvas grab off the preview, which is downscaled
- ☐ Three-step wizard: frontal → left 60° → right 60°, one tap each, auto-advance
- ☐ **Ghost overlay** — fetch the previous session's frame for the current pose, composite at ~35% opacity over the live view
- ☐ Tap-to-focus, defaulted to the region that pose owns rather than frame center
- ☐ `applyConstraints` for manual exposure, fixed white balance, torch — and log what was actually granted, because iOS will grant almost none of it
- ☐ Covariate row after the third shot: sleep, stress, workout, washed, new product, free-text note
- ☐ Automatic QA on upload: blur (variance of Laplacian), exposure clipping, fiducial found — reject and re-prompt on the spot, not three weeks later
- ☐ Webcam fallback script writing to the same paths, tagged `device` so those sessions are never silently mixed in
- ☐ Migrate your manually-shot sessions into the database
- ☒ GATE — a full session in under 60 seconds, done seven days running

PHASE 2 Labeling

WEEK 2–6 · THE BULK OF THE WORK

- ☐ Finish `RUBRIC.md` with real example crops from your own photos, including the borderline cases
- ☐ Build the labeling tool: canvas view, click to place a marker, scroll to size its radius, number keys 1–5 for class, undo, next/prev
- ☐ `labels` table storing center and radius in image pixels; convert to boxes at train time
- ☐ Label the first ~100 images entirely by hand
- ☒ GATE – re-label 20 images from two weeks earlier without looking at the originals. Agreement below ~0.75 means fix the rubric, not the model
- ☐ Train a weak model on those 100 and wire up pre-labeling
- ☐ Add correct-mode to the tool: accept, nudge, delete, add. This is where the 3–5× speedup comes from
- ☐ Label to 300–600 images total, in batches, retraining the pre-labeler as you go

PHASE 3 Canonical space

WEEK 3–4

- ☐ `calib.py` — ArUco detection, px/mm scale, color normalization against the patch
- ☐ Pick and **freeze three atlas images**, one per pose. Never change them; changing one invalidates every historical coordinate under it
- ☐ `register.py` — landmark coarse alignment → feature match to the pose atlas → similarity, then homography, then TPS if residuals warrant → store transform and residual stats
- ☐ Reject-don't-average: residual over threshold sets `reg_status = unregistered` and prompts a re-shoot
- ☐ `regions.py` — draw the six region polygons once in canonical coordinates, each tagged with its owning pose
- ☐ Implement the daily cross-pose check: map mid-cheek detections from both poses into a shared head frame, log the disagreement
- ☐ Blink test — warp three different days into canonical space and flip between them. Moles must not move

PHASE 4 Detector

WEEK 5–7

- ☐ Check whether **ACNE04** is still publicly available, and read its license before using it
- ☐ Tiling: 640 px tiles at ~20% overlap, with label remapping into tile coordinates
- ☐ **Date-based split function.** Contiguous held-out blocks of days — write this before the training script, not after
- ☐ Train YOLOv8n/v11n or RT-DETR — one model across all poses, with pose as an input feature, not three per-pose models splitting your data
- ☐ Augmentation: heavy color jitter and lighting simulation, mild rotation, flips. No aggressive geometric warps — they distort lesion morphology, which is your signal
- ☐ Sliced inference across tiles with NMS merge at the seams
- ☐ Metrics: precision/recall at 2 mm center distance, count MAE per region, day-to-day change error
- ☐ Tune the confidence threshold toward precision — a false positive becomes a phantom lesion in your history
- ☐ GATE — precision above ~0.8 at recall above ~0.7, on held-out days

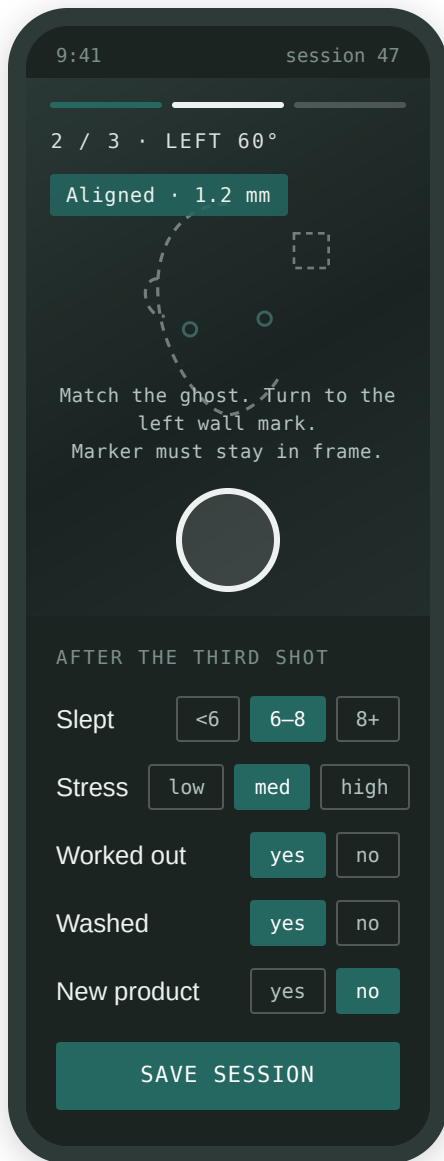
PHASE 5 Lesion tracking

WEEK 7–8

- ☐ Project every *owned* detection into canonical mm coordinates
- ☐ Hungarian matching against active lesions' last known positions — not against yesterday's frame
- ☐ Gate matches at ~4 mm
- ☐ Lifecycle state machine: born on recurrence in 2 of 3 consecutive sessions; resolved after 3 consecutive absences, backdated to the last sighting
- ☐ `lesions` and `observations` tables, with class recorded per observation
- ☐ Manual audit — pick 10 tracked lesions and verify by eye across their whole span. Tracking bugs are invisible in aggregate and obvious in individual traces

- ☐ Active count over time, stacked by region, with a 7-day rolling mean
 - ☐ Canonical density heatmap of lesion-*days*, as small multiples by month
 - ☐ Lesion swimlanes — one bar per lesion, birth to resolution, colored by class
 - ☐ Duration histogram
 - ☐ Wrap in Streamlit or a static Plotly page. Don't build a web app; you're the only user
-

Daily capture



Example state, not real data. The dashed outline is the previous session's frame; teal circles are its detected landmarks.

What each element is doing

- **The ghost is the whole design.** Everything else is chrome around it. You match your face to a translucent copy of yesterday's frame, which is why the yaw angle stays reproducible without a jig.
- **Live alignment readout** — run Face Mesh on the preview stream, compare landmark positions to the ghost's, show the residual in mm. Turns green under ~2 mm. This is the single feature that will most improve your registration quality, because it fixes the problem before the shutter fires instead of rejecting the shot afterward.
- **Three dashes, not a step counter.** The session is short enough that you should never have to read a number to know where you are.
- **No text entry anywhere.** Covariates are taps with three options each. A free-text note field can exist, but it must never be required — the moment a keyboard opens, the 60-second budget is gone.
- **Reject on the spot.** If blur or exposure QA fails, re-prompt that pose immediately. A bad frame you discover in week 6 is a lost day.

BUILD ORDER

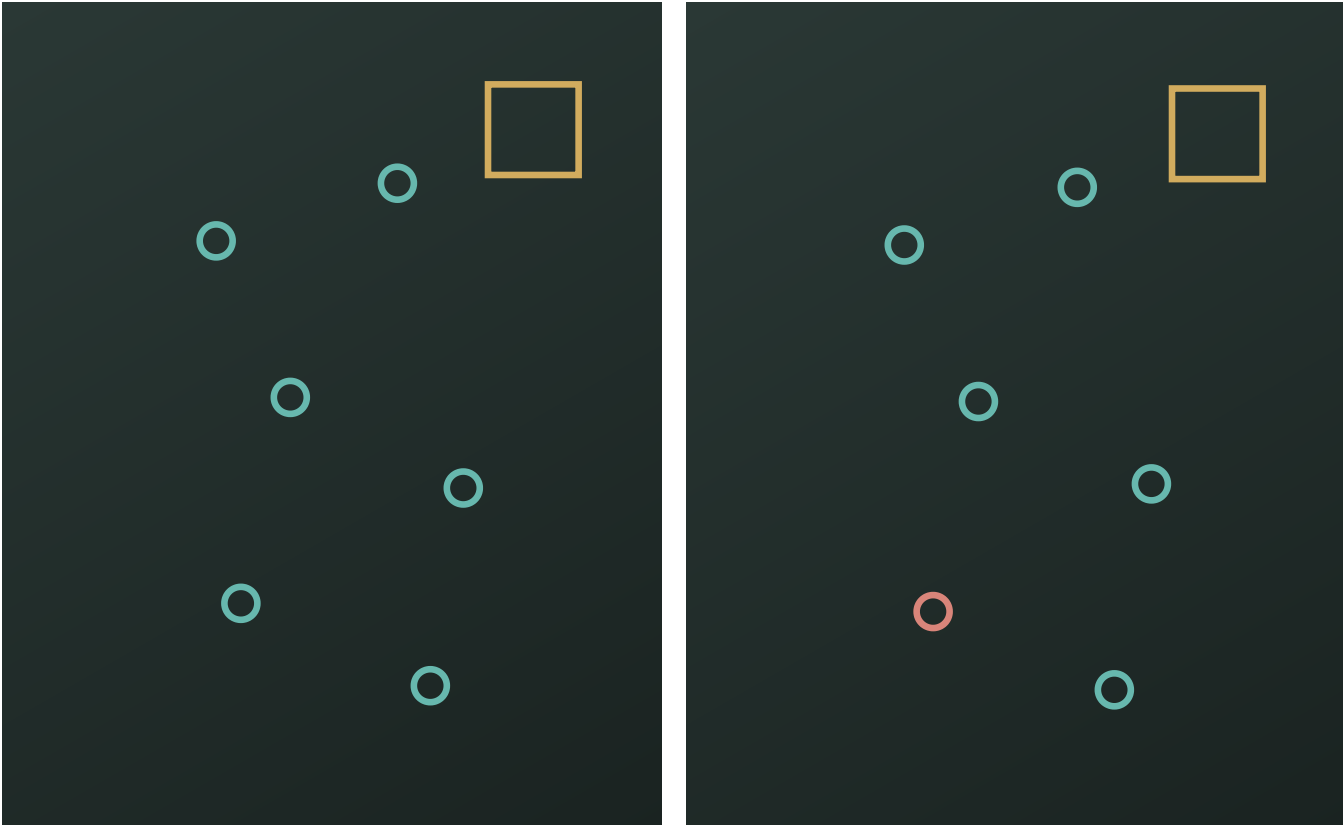
Wizard and upload first, ghost overlay second, live alignment readout last. The first two make the app usable; the third makes it good, and it needs Face Mesh running on a video stream, which is the fiddliest part.

Calibration & registration review

● ● ● registration review – left_60 – 2026-09-14

ATLAS · FROZEN 2026-09-12

TODAY · WARPED TO ATLAS



MEDIAN ERR 0.9 mm	P95 ERR 2.1 mm	MATCHES 412	TRANSFORM TPS	SCALE 26.4 px/mm
FACE MESH OK				

ACCEPT

BLINK COMPARE

FLAG FOR RE-SHOOT

TRY LOFTR

Example figures. Teal circles are matched control points; the red one exceeded the residual threshold.

Notes

- **One screen, two jobs.** In phase 0 you drive it manually to compare methods and lock the angle. In phase 3 it becomes the review queue for sessions that failed automatic QA — same view, same numbers.
- **Everything in millimetres.** Pixels are meaningless across sessions once your standing distance varies by a couple of centimetres. The px/mm readout comes from the fiducial and is your check that scale normalization worked.
- **Blink compare is the honest test.** Numbers can look fine while the warp is subtly wrong. Flipping between atlas and warped image at ~2 Hz makes any real misregistration jump out immediately — moles will visibly jitter.
- **"Flag for re-shoot" has to be one click** and has to be available the same day, while re-shooting is still possible.

INTERFACE 3 • PHASE 2

Labeling

● ● ● label - 2026-09-21 · left_60 · tile 4/12

☒ papule	1
☐ pustule	2
☒ comedone	3
☒ mark	4
☒ other	5

scroll - radius

space - next

z - undo

x - delete

a - accept all

? - rubric

Batch 3 · image 112 / 180 38 min

Example labels, not real detections. The white marker is the one currently selected; the dashed one is a model pre-label awaiting confirmation.

Notes

- **Optimize for keystrokes, not looks.** You'll place several thousand markers. Every extra click multiplies by 3,000. Click places, scroll sizes, a number key classifies, space advances — hand never leaves the mouse and the number row.
- **Label one tile at a time, not the whole frame.** The detector trains on 640 px tiles anyway, and labeling at the zoom the model sees is both faster and more consistent than hunting across a full face.
- **Show the radius in millimetres** while sizing. Your rubric has a minimum size in mm; the tool should make it impossible to violate it accidentally.
- **Pre-labels are visually distinct** (dashed until confirmed), so accepting is deliberate. Once the model is decent you'll be tempted to accept-all and skim — that's how a model starts training on its own mistakes.
- **Keep the rubric one key away.** Borderline cases are constant; a two-second lookup is the difference between a consistent protocol and drift.
- **Track time per batch.** If it isn't falling once pre-labeling is on, the pre-labeler isn't good enough to be worth the correction overhead.

The one thing on this page that is genuinely urgent is starting daily capture. Everything else can slip a week without consequence; a missed session is gone permanently.